

PaliGemma · arXiv 2407.07726 · Sep 16

Scientific Reviewer

seat 1 · Kanta · 8+3

Claim · strengths · weaknesses · 10 actionable asks

Paper club · VLA Architectures · dark academic

Claim under review

**A carefully trained <3B open base VLM
(SigLIP-So400m + Gemma-2B, Prefix-LM, unfrozen Stage1,
resolution Stage2) transfers broadly across ~40 tasks
and is a better research object than chat-tuned VLMs.**

Not a chat demo. Transfer-first base. Open weights load-bearing for robot FMs.

Review frame

Judge as engineering + ablation contribution with open base checkpoints.
Demand boundary language on PaLI-X / PaLM-E comparisons and Stage1 opacity.

Strengths

Why this paper earns attention as a research object

Base vs IT

Clear product thesis:
transfer base, not
chat demo — rare honesty

PaLI scaling

Coherent SigLIP+Gemma
line thesis matching
larger PaLI-family style

Deep ablations

Stage1 length, mask/loss,
freeze/reset, connector,
encoder, res, few-shot

Contamination hygiene

Near-dupe removal;
task prefixes; no commercial
VLM synthetic IT in pretrain

Loc / seg API

1024-bin locs + 128
VQVAE mask tokens —
structured grounding

Systems transparency

MFU, tok/s, Stage
walltimes, App J transfer
recipes, open weights

Weaknesses

Private Stage1

Mixture blocks independent
recipe reproduction at
paper scale — open weights
≠ open science

Material limits on recipe claims and robotics over-read

Table 1 self-vs-res

Mostly self across 224/
448/896 — not matched
public bake-off vs
Idefics2 / QwenVL / InternVL

Macro-average

Metric mixing can obscure
per-family performance;
SciCap 60% regret under
defaults is a red flag

Stage3 ≠ robotics

Per-task FT ≠ OOD
embodied transfer;
don't invent BEHAVIOR
wins from Table 1

Square resize

May hurt embodiment
cameras / aspect ratios;
no action / memory / 3D
objectives

TPU-centric

No GPU/LoRA memory
tables in paper; uneven
seeds; LoRA = Empiricist
extension

10 actionable improvements

Say these aloud · THE talk figure · gift to authors / Empiricist

1 Public Stage1 surrogate mixture / sampling weights

2 Fixed public baseline panel (Idefics2 / QwenVL / InternVL) under SAME transfer recipe

3 Per-family normalized scores instead of opaque macro-average

4 Robotics probes: egocentric VQA, affordance pointing, multi-cam correspondence

5 Aspect-ratio / camera-model ablations

6 loc/seg calibration + constrained decoding analysis

7 Continuous ViT:LLM LR-ratio sweeps (not binary freeze)

8 Preregister resolution-sensitive tasks; corrected win rates

9 One-command public Table1 harness from released ckpts

10 Negative-results appendix (e.g. SciCap multitask collapse)

PaliGemma: A versatile 3B VLM for transfer

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PaliGemma is an open Vision-Language Model (VLM) that is based on the SigLIP-So400m vision encoder and the Gemma-2B language model. It is trained to be a versatile and broadly knowledgeable base model that is effective to transfer. It achieves strong performance on a wide variety of open-world tasks. We evaluate PaliGemma on almost 40 diverse tasks including standard VLM benchmarks, but also more specialized tasks such as remote-sensing and segmentation.

1. Introduction

PaliGemma is an open model, continuing the line of PaLI vision-language models in a combination with the Gemma family of language models.

PaLI is a series of state-of-the-art vision-language models, starting with the first PaLI [23] showing promising scaling results up to 17 B, using classification pretrained ViT [131] and mT5 [126] language model. PaLI-X [24] and PaLM-E [36] then pushed this further, combining ViT-22 B [29] and a 32 B UL2 [104] language model or the 540 B PaLM [28] language model, respectively, and getting further increased performance on vision-language tasks, albeit saturating performance on standard image classification and retrieval tasks. Finally, PaLI-3 [25] demonstrates that through better pretraining with SigLIP [133] and more careful multimodal data curation, a 2 B vision and 3 B language model (*i.e.* a 5 B vision-language model) matches the 10x larger PaLI-X and 100x larger PaLM-E across most benchmarks.

PaliGemma continues this trend, combining the 400 M SigLIP and the 2 B Gemma models [82] into a sub-3 B VLM that still maintains performance comparable to PaLI-X, PaLM-E, and PaLI-3.

Gemma [82] is a family of auto-regressive decoder-only open large language models built from the same research and technology used to create the Gemini [7] models. The models come in different sizes (2 B, 7 B), both pretrained and

instruction fine-tuned. PaliGemma uses the 2 B pretrained version.

The main goal of our work is to provide a versatile base VLM. Hence, we show that it reaches state-of-the-art results not only on standard COCO captions, VQAv2, InfographicVQA and others, but also on more exotic Remote-Sensing VQA, TallyVQA, several video captioning and QA tasks, as well as referring expression segmentation (see full task list in Appendix B).

2. Related work

Over the course of the past few years, vision-language models have gained considerable importance in computer vision. The first generation, spearheaded by CLIP [94] and ALIGN [49] by scaling up ConVIRT [135] and VirTex [32], is an extension of large-scale classification pretraining [55, 131], to leverage all data from the web without the need for onerous human labeling, replacing a fixed and large set of classes by a caption embedding instead. The caption embeddings are mostly obtained using language encoders (similar to BERT [33]) and allow to open up the vocabulary of classification and retrieval tasks. The second generation, akin to T5 [95] in language, is a unification of captioning and question-answering tasks via generative encoder-decoder modeling [27, 111, 120, 138], often backed by the progress in generative language models.

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PaLI traditionally uses an encoder-decoder language model; however all recently publicly released language models are decoder-only Transformers. We opt for the Gemma-2B model, which strikes a good balance between size and performance. Larger language models, such as the popular 7 B or 70 B sizes, are often significantly better at tasks like mathematical reasoning. However, PaLI-3 has shown that across a wide range of vision-language tasks, a well-trained small 5 B model (2 B vision + 3 B language) can attain the same performance as the much larger 55 B PaLI-X (22 B vision + 32 B language) and 562 B PaLM-E (22 B vision + 540 B language), including tasks such as ScienceQA. With PaliGemma we continue this push for smaller models and show that we can keep the same performance with less than 3 B total parameters.

3.2.2. Stage1: Multimodal pretraining

In this stage, we combine the unimodal models as explained in Section 3.1 and train the whole model on a broad mixture of large-scale vision-language tasks. Contrary to most recent VLMs, our core goal is to train a base model that fine-tunes well to a wide range of tasks, not merely to align the modalities. Intuitively, we want a mix of tasks which force the model to acquire a broad range of “skills”, regardless of the task’s user (or benchmark) friendliness out of the box. More on this in Section 3.2.5.

It is common practice, also followed by previous PaLI versions, to keep the image encoder frozen during the first multimodal pretraining stage. This is partially due to findings as in LiT [132] reporting multimodal tuning of pre-trained image encoders degrading their representations. However, more recent work such as CapPa [110] and LocCa [115] have shown that captioning and other harder-to-learn tasks can provide valuable signal to image encoders, allowing them to learn spatial and relational understanding capabilities which contrastive models like CLIP or SigLIP typically lack. Hence, again in the spirit of learning more skills during pre-training, we depart from common practice and do not freeze the image encoder. However, the challenges outlined in LiT remain. In order to

avoid destructive supervision signal from the initially unaligned language model, we use a slow linear warm-up for the image encoder’s learning-rate (Figure 3), which ensures that the image encoder’s quality is not deteriorated from the initially misaligned gradients coming through the LLM.

We train Stage1 at resolution 224px (hence, $N_{\text{img}} = 256$ image tokens) and sequence length $N_{\text{txt}} = 128$ for a total of 1 billion examples. While we provide an ablation in Section 5.1 showing that a 10x to 30x shorter Stage1 still provides good results on popular benchmarks, we wish to imbue as much visual knowledge to the base model as possible, and cover a broad set of concepts, cultures, and languages [17, 37, 68, 85, 92, 93, 136].

3.2.3. Stage2: Resolution increase

The model resulting from Stage1 is already a useful base model for many tasks (see example images in Appendix B). However, it only understands images at 224×224 pixel resolution, which is too small for several tasks. For instance, detection and segmentation of smaller objects, and tasks related to reading smaller texts such as charts, infographics, or documents, all strongly benefit from higher resolution (see Table 1). Hence, we train two further model checkpoints for increased resolution, first to 448×448 and then to 896×896 pixel resolution.

Since stage1 took care of providing the model with a broad set of knowledge and skill, stage2 can focus on extending the model’s ability to parse higher-resolution images. We thus run Stage2 with fewer total examples, while increasing the cost and information density of each example. For resolution 448, we train for an additional 50 M examples, and for resolution 896, we add another 10 M examples.

For simplicity, Stage2 consists of the exact same mixture of tasks and datasets as Stage1, but with significantly increased sampling of tasks that require high resolution. Additionally, these up-weighted tasks all can be modified to provide much longer suffix sequence lengths. For instance, for OCR tasks, we can simply request the model

Score intuition

Strong open-model + ablation contribution

Landmark if data opacity is acknowledged

Methods venues should demand public mixture + matched baselines

Lean accept

Engineering & ablation paper with open base ckpts. Useful practitioner transfer recipes.

Require

Boundary language vs 100x models; fuller zero-shot protocol disclosure; GPU-oriented resource appendix.

Pressure-test

Empiricist LoRA / freeze / resolution follow-ups on real clusters — right way to stress transfer.

Reviewer close

Base claim real · Stage1 opacity real · 10 asks are the gift